

# Prudential SG Policy, Claims & Agent Book

## Data Quality & Exploratory Data Analysis — Business Report

Step 1 deliverable: Data Collection & EDA covering structural profiling, data-quality diagnostics, statistical exploration and visual analysis of the raw extract **prudential\_sg\_raw.csv** (50,000 records, 65 fields).

|                    |                                               |
|--------------------|-----------------------------------------------|
| Prepared for       | Analytics / Data Science stakeholders         |
| Source file        | prudential_sg_raw.csv                         |
| Records examined   | 50,000 rows × 65 columns                      |
| Report date        | 10 September 2026                             |
| Companion artefact | prudential_edu.py (reproducible EDA pipeline) |

*Note on the data: this extract shows patterns typical of a consolidated, multi-source-system export awaiting a formal cleansing layer — mixed field formats, placeholder codes, and a handful of fixed sentinel/error values. All findings in this report describe this extract only and should be validated with the relevant system owners before being used for external reporting.*

## Executive Summary

**50,000**

Raw rows

**2,500**

Duplicate rows removed  
(exact + fuzzy)

**9.5%**

Overall policy lapse rate

**11.8%**

Claim decline rate

The extract contains 50,000 policy-level records across 65 fields spanning quote dates from January 2023 to mid-2026, combining customer, policy, claims, and servicing-agent attributes in a single wide table. Before any of it can be reported on or modelled, it needs a deliberate cleaning pass: 2,500 rows (5.0%) are duplicates of another row — 1,500 byte-for-byte exact duplicates plus a further 1,000 disguised as new records by a formatting quirk (Section 2.2) — and the vast majority of fields mix multiple raw formats (currency notations, date formats, and up to twelve different spellings of “Yes/No” flags) alongside placeholder text (‘unknown’, ‘?’, ‘-’) standing in for missing values.

### Headline findings

- **2,500 rows (5.0%) were duplicates** — 1,500 byte-for-byte exact repeats plus a further 1,000 policies re-entered a second time with a formatting quirk (the customer's name losing its spacing) that let them slip past exact-match de-duplication entirely. Verified three independent ways before removal (Section 2.2).
- **Data quality is the main story here, not a footnote.** Every numeric field examined (income, premium, sum assured, claim amount) was stored as free-format text; two fields (**claim\_amount\_sgd**, **claim\_tat\_days**) contain fixed sentinel/error codes (e.g. a repeated 99,999,999.99 claim amount) that inflate their raw averages by 20–50× until removed.
- **lapse\_flag is fully derived from policy\_status** (True exactly when status is Lapsed, Surrendered or Cancelled). The two fields cannot be used as independent predictors of one another, and the business should confirm whether “lapse” is meant to bundle in surrenders and cancellations.
- **claim\_decline\_reason is populated in exactly the rows where a claim was declined** (3,349 of 3,349) — a perfect target leak if ever used as a model feature rather than an output.
- Roughly 40–84% “missing” on claim, rider and agent fields is mostly **structural, not lost data**: those fields simply don't apply when no claim was filed, no rider was attached, or the policy was sold outside the Agency/Broker channel.
- Across the book, the **overall lapse rate is 9.5%** and **60% of policies carry at least one claim**, of which **11.8% are declined**. Lapse risk is highest for Bancassurance-sold policies (11.3%) and lowest for Broker-sold policies (7.9%).
- Demographic fields (age, income) show **essentially no linear correlation** with premium, sum assured, or lapse in this extract — worth validating against the actual underwriting/pricing logic, since it is atypical for an insurance book.

### What this means for next steps

Section 5 sets out a concrete cleaning checklist and modelling-readiness notes. In short: standardise formats and flag encodings, null out the sentinel/placeholder values identified in Section 2, resolve the lapse\_flag / policy\_status duplication with the business, and treat the claims/rider/agent blocks as conditionally-populated rather than randomly missing when choosing an imputation strategy.

# 1. Dataset Overview

The extract is a wide, denormalised snapshot combining four source systems (roughly equal shares from CRM, Policy Administration, Agency Management, and Claims): one row is intended to represent one policy record, carrying customer demographics, policy/product terms, claims history, and the servicing agent's profile side by side.

| Metric                                | Value                                                                   |
|---------------------------------------|-------------------------------------------------------------------------|
| Rows (raw)                            | 50,000                                                                  |
| Columns                               | 65                                                                      |
| Memory footprint (raw, in-memory)     | ~134 MB                                                                 |
| Duplicate rows (exact + fuzzy)        | 2,500 (5.0%) — removed for analysis; see Section 2.2                    |
| Rows after de-duplication             | 47,500                                                                  |
| Source systems                        | CRM, Policy Administration, Agency Management, Claims (~25% share each) |
| Date range covered (quote_date)       | Jan 2023 – Jun 2026 (sparse/likely erroneous tail beyond)               |
| Primary key                           | record_id (unique after de-duplication)                                 |
| Recommended primary target            | lapse_flag (binary; defined for all rows)                               |
| Secondary targets (conditional scope) | claim_status, fraud_risk_score / sod_breach_flag, persistency_13m_flag  |

## Candidate unique identifiers

| Field       | Unique values | Non-null rows | Role                                              |
|-------------|---------------|---------------|---------------------------------------------------|
| record_id   | 47,500        | 47,500        | Primary key of the final, de-duplicated dataset   |
| policy_id   | 47,259        | 49,840        | One customer can hold several policies            |
| customer_id | 28,135        | 49,778        | Average ~1.75 policies per customer               |
| claim_id    | 28,366        | 29,898        | Populated only when a claim exists (~60% of rows) |
| agent_id    | 1,372         | 26,256        | Populated only for Agency/Broker-sold policies    |

## Target variable identification

This is a source extract, not a pre-built model table, so “the” target depends on the question being asked. The table below ranks the candidates considered and the reasoning for recommending **lapse\_flag** as the primary target for this report's bivariate cuts.

| Candidate target                   | Business use case                  | Coverage                             |
|------------------------------------|------------------------------------|--------------------------------------|
| lapse_flag (recommended)           | Policy lapse / churn prediction    | 100% of rows                         |
| policy_status                      | Multi-class lifecycle state        | 100% (but redundant with lapse_flag) |
| claim_status                       | Claims triage / decline prediction | ~60% (only rows with a claim)        |
| fraud_risk_score / sod_breach_flag | Fraud & compliance monitoring      | ~60% (claims only)                   |
| persistency_13m_flag               | 13th-month persistency KPI         | 100% of rows                         |
| agent_attrition_risk               | Agent workforce analytics          | ~52% (Agency/Broker only)            |

## 2. Data Quality Assessment

This section documents quality issues found in the **raw** extract, before any cleaning was applied — this is the evidence base for the cleaning pipeline in the companion script and the checklist in Section 5.

### 2.1 Missing values — declared vs. hidden

Raw NaNs alone understate the problem: placeholder tokens such as '-' '?' 'unknown' and blank strings appear as “valid” text in roughly 20 categorical columns and should be folded into the missing count. For example, **gender** shows only 172 raw NaNs but a further ~350 rows use one of nine junk placeholders; **distribution\_channel** and **policy\_status** show the same pattern.

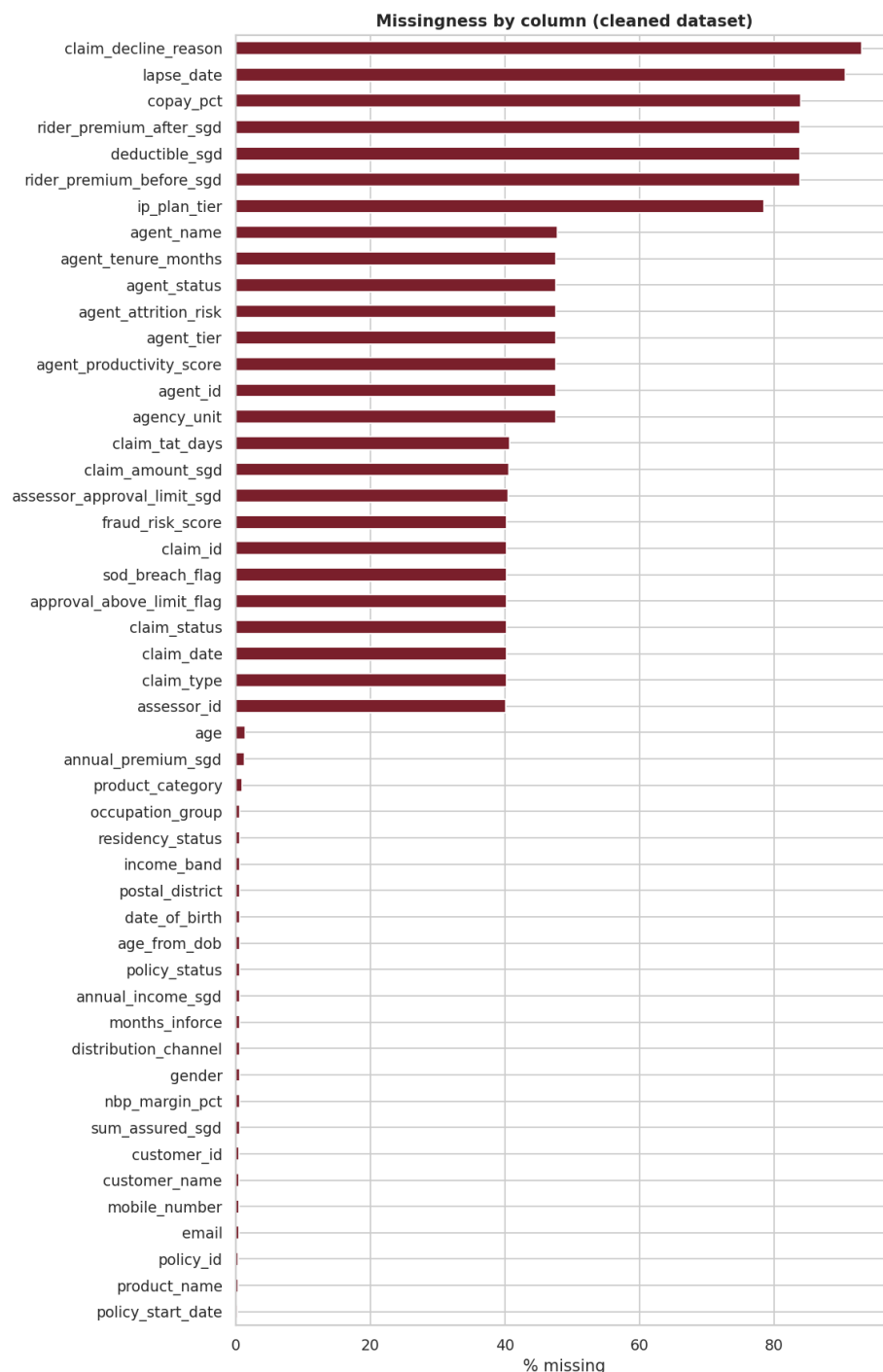


Figure 2.1 — Missingness by column (cleaned dataset; % of rows). The top block (claim, rider, agent fields) is mostly structural non-applicability, not lost data — see §2.6.

## 2.2 Duplicate records

**Exact duplicates.** 1,500 rows (3.0% of the file) are byte-for-byte duplicates of another row in the extract, including identical **record\_id**, **last\_updated\_ts** and every other field — this points to a repeat extraction/append issue in the source pipeline rather than two genuinely separate events. These were dropped (keeping the first occurrence) before any statistics or charts in this report were produced.

**Fuzzy duplicates.** Exact matching only catches rows identical in every field. Grouping the remaining 48,500 rows on (policy\_id + customer\_name with whitespace and case stripped) surfaces a second, subtler pattern: 1,000 policies genuinely re-entered a second time where the customer's name lost its internal spacing on the repeat (e.g. 'Jonathan Bennett' vs 'JonathanBennett') — product, premium and quote date identical in every other respect. This was confirmed three independent ways before being treated as a genuine duplicate: matching on policy\_id, matching on customer name + date of birth + policy fingerprint (product/premium/quote date) while ignoring policy\_id entirely, and matching on e-mail address + the same policy fingerprint — all three landing on exactly 1,000 pairs. A looser name-similarity threshold was also tested and rejected: it starts matching genuinely different customers who simply share a first name and a common surname (e.g. 'Rachel Teo' vs 'Rachel Neo', confirmed as two different policies with different premiums and quote dates), so the rule was kept deliberately narrow.

In total, **2,500 rows (5.0% of the raw file) are duplicates** and were removed before any of the statistics, correlations or charts in this report were produced, leaving **47,500 unique policy records**.

## 2.3 Inconsistent formatting

| Field(s)                                            | Formats observed                                                                                                                   |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| gender                                              | 15 spellings: M/male/MALE/F/Femal/female/blank/'?' ...                                                                             |
| distribution_channel                                | 26 variants: case differences, leading spaces, and typos ('Bncssurnc'→Bancassurance, 'Digitl'→Digital)                             |
| premium / income / sum assured / APE / claim amount | Mixed currency notation: plain number, '\$\$4,521.97', 'SGD 1,805.01', '\$ 4521.97', with/without thousands separators             |
| nbp_margin_pct                                      | Mixed scale: plain fraction (0.0613) vs. percentage string ('20.39%')                                                              |
| date fields (5 columns)                             | 5 formats mixed in the same column (DD-MM-YYYY, DD.MM.YYYY, 'Month DD, YYYY', DD-Mon-YY) plus an Excel-epoch sentinel (1899-01-01) |
| age / months_inforce                                | Numeric fields stored as text, interleaved with placeholder text                                                                   |
| 8 boolean flag columns                              | Up to 12 encodings of the same Yes/No: Y/N, Yes/No, yes/no, TRUE/FALSE, T/F, 1/0                                                   |
| customer_name / agent_name                          | Leading / trailing / doubled whitespace                                                                                            |

## 2.4 Outliers & invalid sentinel values

Beyond organic outliers, several fields carry **fixed sentinel codes** that are clearly injected error markers rather than real values — these need to be nulled out, not winsorised, since they are not part of the true distribution:

| Field              | Issue                                                                  | Rows affected |
|--------------------|------------------------------------------------------------------------|---------------|
| claim_amount_sgd   | Fixed sentinel 99,999,999.99 (system error code)                       | 72            |
| claim_amount_sgd   | Fixed sentinel -\$15,000 (invalid negative)                            | 84            |
| claim_tat_days     | Fixed sentinel of 4,000 days (real max is 102 days)                    | 76            |
| annual_premium_sgd | Organic negative values across a wide range (data error)               | ~330          |
| copay_pct          | Out of the valid 0–100% range (min –10, max 999)                       | ≈few dozen    |
| age                | Below 18 or above 100 years old                                        | ~250          |
| policy_term_years  | Value of 99 likely a 'Whole-of-Life' business code, not a literal term | 7,108         |

The two claim\_amount\_sgd sentinels alone inflate the raw column average from a realistic ~SGD 8,700 to an apparent ~SGD 257,000 (30× distortion) until removed — any report built on the unclean average would materially misstate typical claim severity.

## 2.5 Class imbalance & cardinality

| Indicator                   | True rate | Imbalance         |
|-----------------------------|-----------|-------------------|
| lapse_flag (primary target) | 9.5%      | ~9.6 : 1          |
| persistency_13m_flag        | 94.5%     | ~17 : 1 (inverse) |
| complaint_flag              | 3.4%      | ~28 : 1           |
| disputed_flag               | 0.9%      | ~106 : 1          |
| chronic_condition_flag      | 18.9%     | ~4.3 : 1          |
| rider_attached_flag         | 16.1%     | ~5.2 : 1          |

Cardinality is as expected for a table of this kind: identifiers (record/policy/customer/claim/email) are near-unique as intended; demographic and product fields sit in the 2–32 category range; and **approval\_above\_limit\_flag** is near-constant (≈97% 'No' among claim rows), giving it limited standalone predictive value.

## 2.6 Leakage risk

Four structural relationships were confirmed statistically and should shape any modelling scope built on this data:

- **claim\_decline\_reason ↔ claim\_status.** The reason field is populated in exactly the 3,349 rows where claim\_status = 'Declined' and nowhere else — a perfect 1:1 map. It is an explanation of the outcome, not an independent predictor of it.
- **lapse\_flag ↔ policy\_status.** lapse\_flag is True if and only if policy\_status is Lapsed, Surrendered, or Cancelled (confirmed on all 47,500 rows bar 23 edge cases). Using both as separate model inputs double-counts the same event.
- **Post-decision claim fields.** claim\_tat\_days, fraud\_risk\_score, assessor\_id/limit, and sod\_breach\_flag are only known once a claim has been assessed. Using them to predict the claim\_status outcome itself (rather than as post-hoc explanatory fields) leaks the future into the past.
- **Structural (not random) missingness.** Rider premiums/copay/deductible are populated exactly when rider\_attached\_flag is true; ip\_plan\_tier is populated exactly for Health & IP products; agent fields are populated almost exclusively (>99%) for Agency/Broker-sold policies. Treat these as 'not applicable' fields to be segmented or flag-encoded, not randomly missing values to impute.

## 2.7 Suspicious / placeholder records

127 rows carry a product\_name of 'ZZ-Retired-Plan' or 'TEMP PRODUCT', and product\_category also includes junk labels ('TBC', 'Misc', 'LEGACY-XX', 'Unknown Product'). These read as test/placeholder entries that leaked into the production extract and are candidates for exclusion pending confirmation with the source team.

## 3. Statistical & Business Insights

All statistics in this section are computed on the **cleaned** 47,500-row dataset (duplicates removed, formats standardised, sentinel values nulled) produced by the companion script.

### 3.1 Descriptive statistics & distribution shape

| Variable                     | Median  | Mean     | Skew  | Shape                                                |
|------------------------------|---------|----------|-------|------------------------------------------------------|
| Age                          | 42 yrs  | 42.6 yrs | 0.17  | Approx. symmetric                                    |
| Annual income (SGD)          | 77,109  | 89,814   | 2.04  | Strongly right-skewed                                |
| Annual premium (SGD)         | 8,285   | 18,010   | 0.90  | Moderately right-skewed                              |
| Sum assured (SGD)            | 442,000 | 726,045  | 1.51  | Strongly right-skewed                                |
| NBP margin                   | 12.2%   | 13.9%    | 0.66  | Moderately right-skewed                              |
| Claim amount (SGD, cleaned)  | 4,957   | 8,712    | 7.05  | Strongly right-skewed (heavy tail from large claims) |
| Customer sentiment (−1 to 1) | 0.40    | 0.38     | −0.53 | Left-skewed, tilted positive                         |
| NPS score (0–10)             | 8       | 7.6      | −0.83 | Left-skewed, tilted positive                         |

The gap between median and mean on income, premium, and sum assured confirms heavy right-skew typical of financial variables — a small number of high-value policies pull the average well above the typical case. Medians are the more representative summary for these fields.

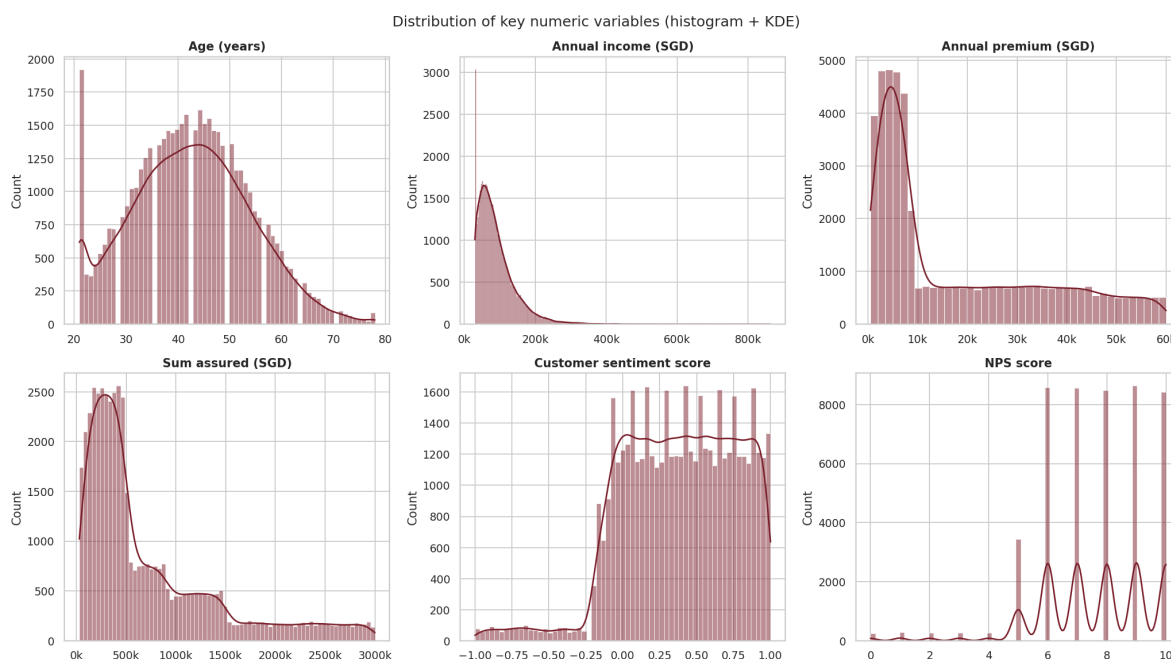


Figure 3.1 — Distribution of key numeric variables (histogram + KDE).

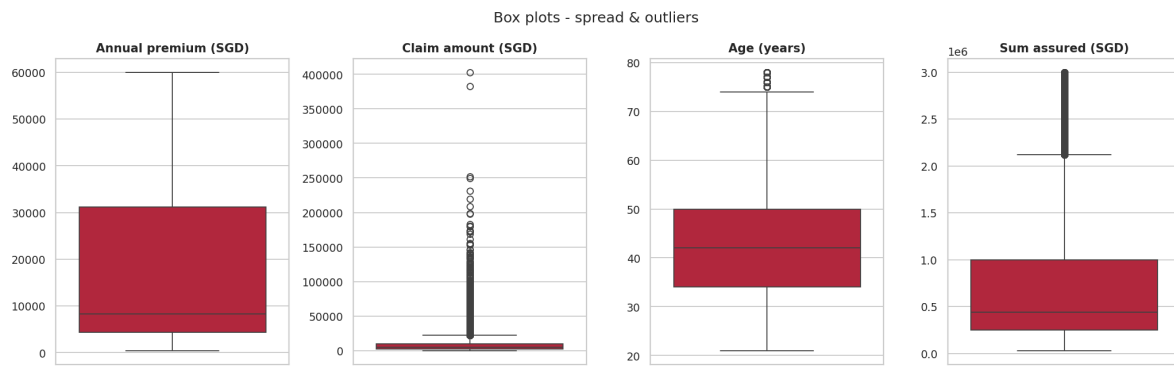


Figure 3.2 — Spread and outliers for premium, claim amount, age and sum assured.

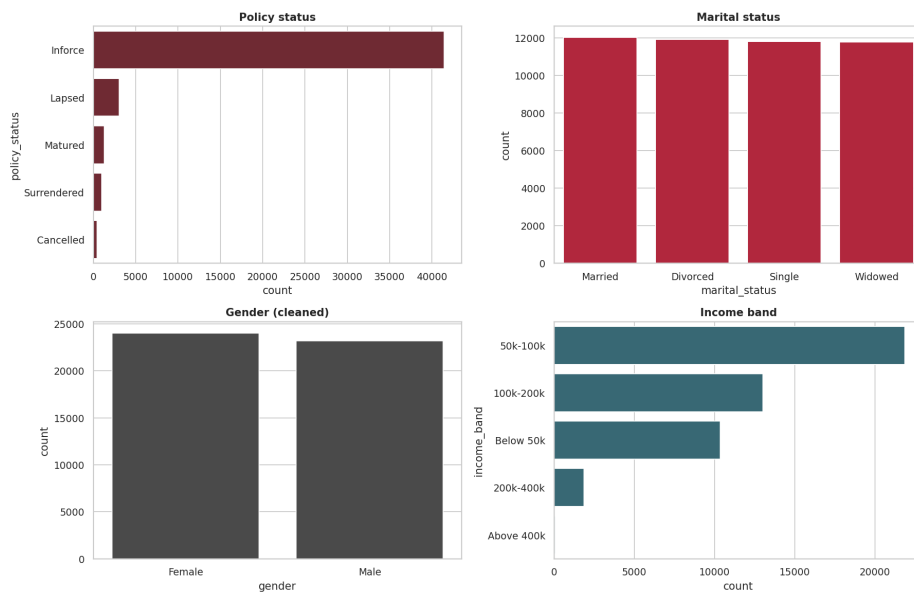


Figure 3.3 — Policy status, marital status, gender and income-band composition of the book.

## 3.2 Correlation between numeric variables

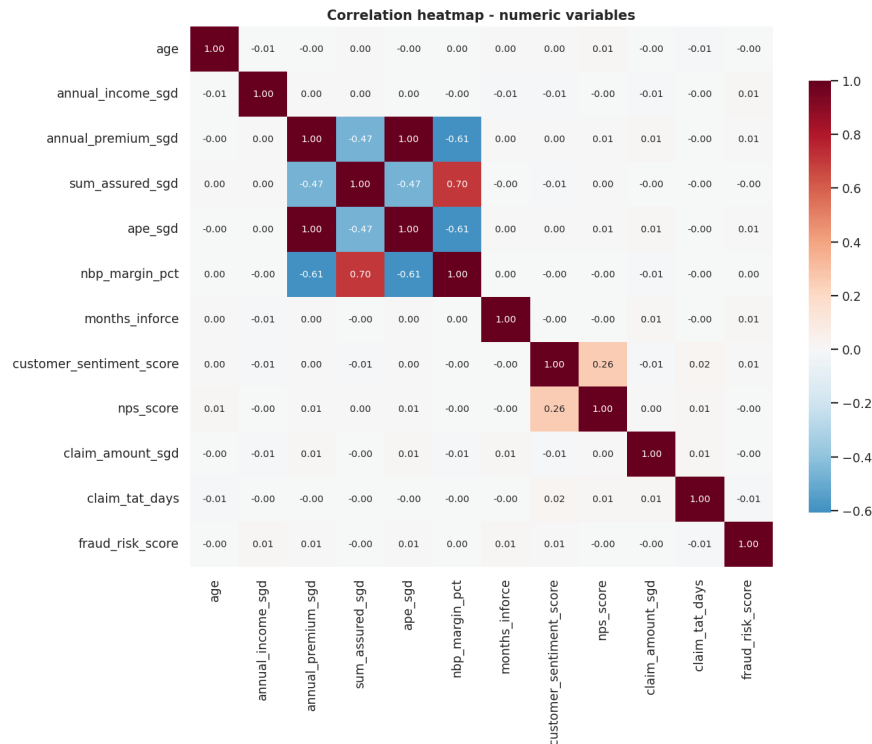


Figure 3.4 — Pearson correlation heatmap, core numeric fields.

**annual\_premium\_sgd and ape\_sgd correlate at 1.00** — the two fields are effectively duplicates for the regular-premium products in this book and should not both be fed into a model as independent features. **nbp\_margin\_pct** correlates moderately with sum assured (0.70) and negatively with premium/APE (−0.61 each, consistent since the two are near-duplicates of one another) — a sensible pricing relationship. Premium/APE also correlate negatively with sum assured itself (−0.47). Outside of the premium/APE/sum-assured/margin cluster, **correlations are close to zero** — notably, age and income show almost no linear relationship with premium, sum assured, or claim amount, which is atypical for an insurance pricing book and worth validating against the real underwriting rules.

## 3.3 Lapse rate by segment (categorical vs. target)

| Distribution channel | Lapse rate | Base (n) |
|----------------------|------------|----------|
| Bancassurance        | 11.3%      | 14,809   |
| Digital              | 10.8%      | 5,743    |
| Agency               | 8.2%       | 21,520   |
| Direct               | 8.4%       | 1,917    |
| Broker               | 7.9%       | 3,240    |

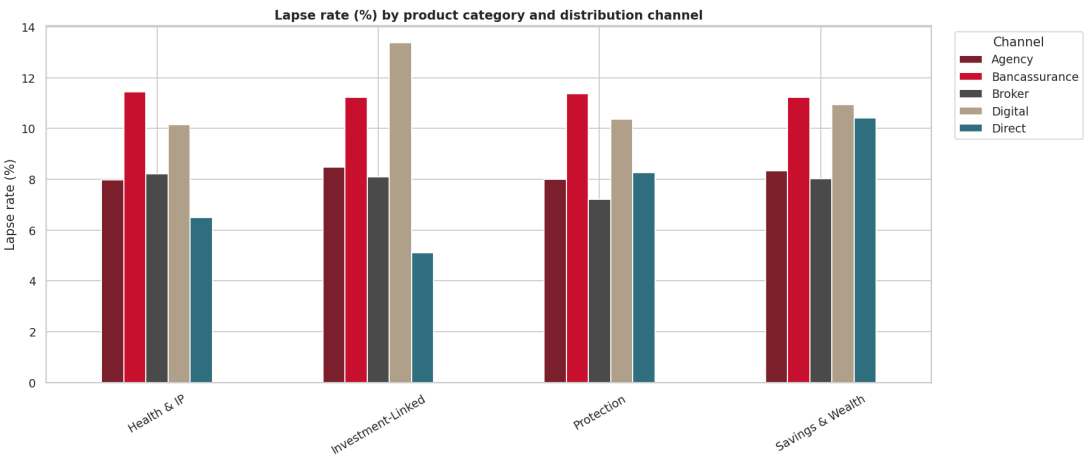


Figure 3.5 — Lapse rate (%) by product category and distribution channel. Bancassurance runs highest across every product line; Broker runs lowest.

3.4 Trend over time

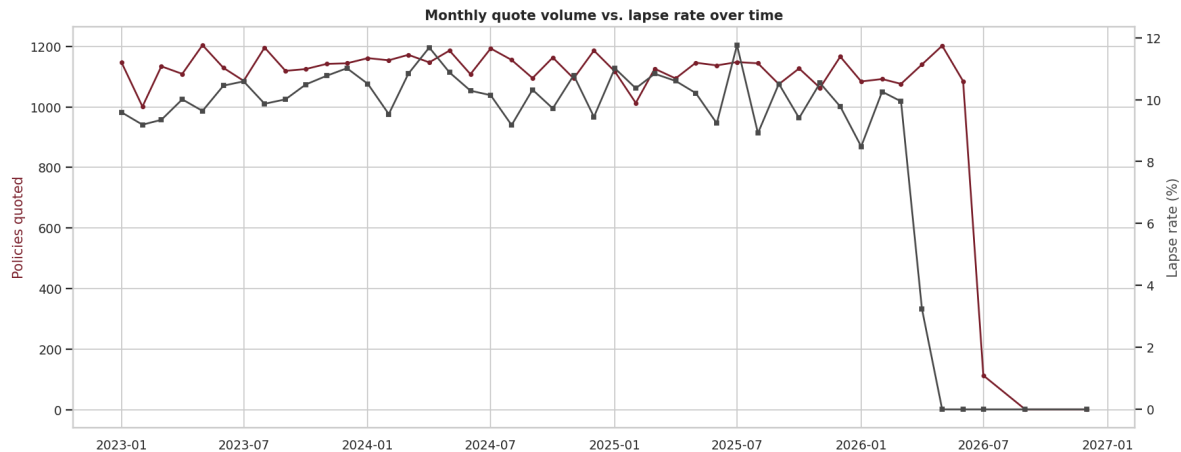


Figure 3.6 — Monthly quote volume (maroon) vs. lapse rate (grey) over time. Volume is broadly flat at ~1,000–1,200 quotes/month from Jan-2023 to mid-2026, tapering sharply thereafter; a single stray quote is dated December 2026, after the report’s reference date, and should be treated as a data error.

3.5 Multivariate view

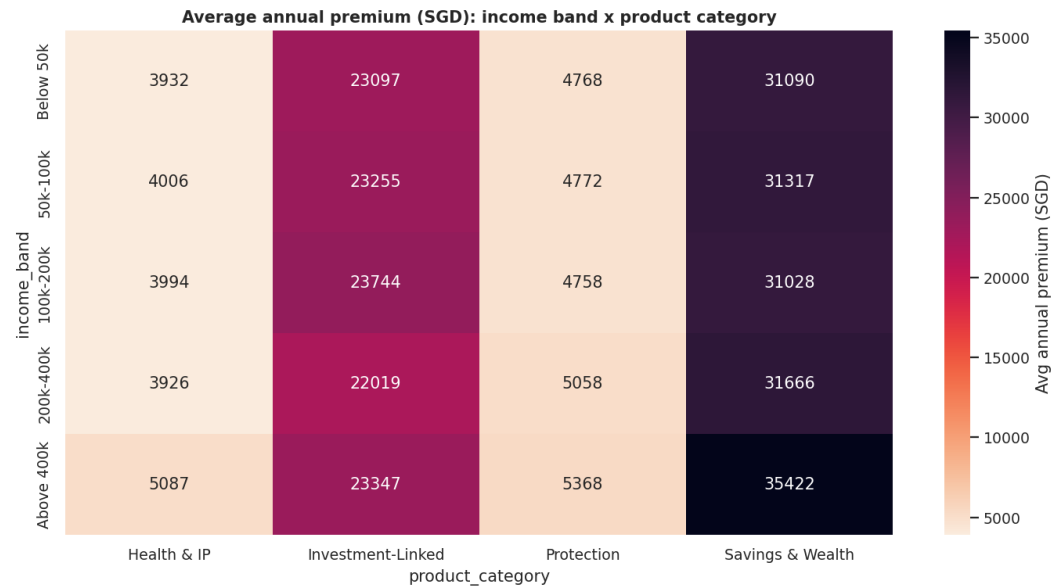


Figure 3.7 — Average annual premium (SGD) by income band and product category. Premium level is driven almost entirely by product category, with essentially no gradient across income bands — consistent with the near-zero income/premium correlation in §3.2.

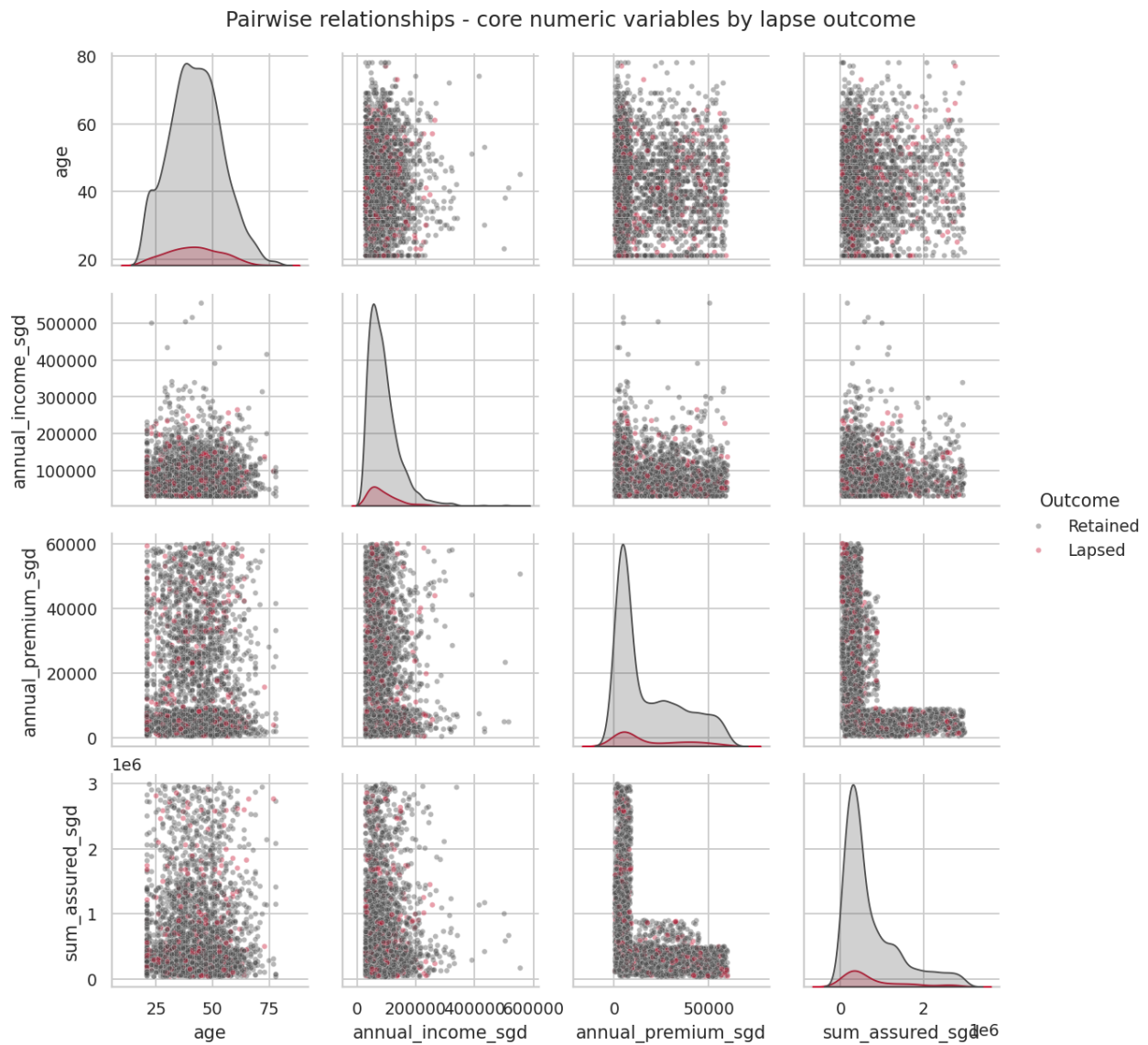


Figure 3.8 — Pairwise relationships between age, income, premium and sum assured, coloured by lapse outcome. No single pair separates lapsed from retained policies visually — lapse risk in this book is not explained by these four variables alone.

## 4. Missing-Data Pattern

The matrix below samples 600 rows to show *where* missingness co-occurs. The claims/rider/agent block (left two-thirds) is missing in large contiguous bands that align with `claim_occurred`, `rider_attached_flag` and `distribution_channel` — visual confirmation that this is conditional/structural missingness rather than random sensor-style dropout. The right-hand columns (demographic & policy fields) show the sparse, scattered pattern typical of genuine data-entry gaps.

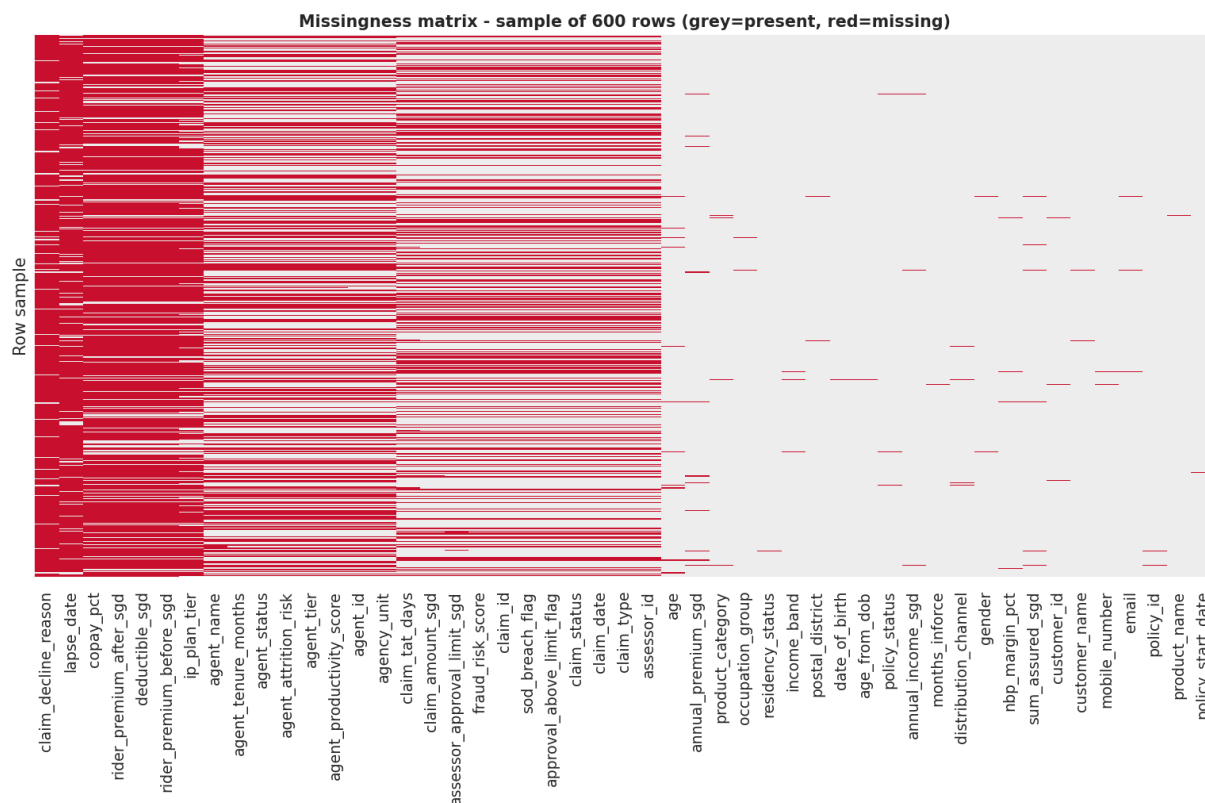


Figure 4.1 — Missingness matrix, 600-row sample (grey = present, red = missing).

## 5. Recommendations & Next Steps

### 5.1 Cleaning checklist before any downstream use

- De-duplicate on full-row match (or record\_id) before any aggregation — already implemented in the companion script.
- Standardise all currency fields by stripping 'S\$', 'SGD', '\$' and thousands separators, then cast to float.
- Standardise nbp\_margin\_pct to a single fraction scale (convert any 'xx.xx%' strings by dividing by 100).
- Map all boolean-style flags (lapse\_flag, rider\_attached\_flag, chronic\_condition\_flag, complaint\_flag, disputed\_flag, sod\_breach\_flag, persistency\_13m\_flag, approval\_above\_limit\_flag) onto a single Y/N or True/False encoding.
- Parse dates with a multi-format fallback (DD-MM-YYYY, DD.MM.YYYY, 'Month DD, YYYY', DD-Mon-YY) and null out the 1899-01-01 sentinel.
- Null out the confirmed sentinel codes: claim\_amount\_sgd = 99,999,999.99 or -\$15,000; claim\_tat\_days = 4,000; copay\_pct outside 0–100%; age outside 18–100; any negative premium.
- Confirm with the source-system owner whether policy\_term\_years = 99 should be recoded as a 'Whole-of-Life' category rather than a numeric term.
- Decide, with the business, whether 'TEMP PRODUCT' / 'ZZ-Retired-Plan' rows (127 total) should be excluded from production reporting.

### 5.2 Modelling-readiness notes

- Treat rider/claims/agent fields as **conditionally structured** data: segment models by channel/product where these fields apply, rather than imputing a single global average across populations where the field is simply not applicable.
- Do not use policy\_status as a feature alongside lapse\_flag, or claim\_decline\_reason as a feature for predicting claim\_status — both are confirmed leakage (§2.6).
- Drop one of annual\_premium\_sgd / ape\_sgd from any regression or tree-based feature set given their 1.00 correlation.
- Given the near-zero linear correlation between age/income and pricing fields, consider non-linear or interaction-based feature engineering before concluding demographics have no pricing relationship in the true source system.

### 5.3 Suggested follow-on analyses

- **Lapse/persistency model** using lapse\_flag or persistency\_13m\_flag as target, segmented by distribution channel given the 3.5-point spread in lapse rate already observed (Bancassurance 11.3% vs. Broker 7.9%).
- **Claims triage / decline-risk model** restricted to claim-level features known at intake (excluding the post-decision fields flagged in §2.6).
- **Fraud & compliance monitoring** using fraud\_risk\_score and sod\_breach\_flag, scoped to the ~60% of rows where a claim exists.
- **Agent productivity & attrition analysis** for the Agency/Broker channel, where agent\_tenure\_months, agent\_productivity\_score and agent\_attrition\_risk are populated.

## Appendix A — Raw Column Dictionary

Full column-level profile of the raw extract (dtype as read from CSV, raw NaN missingness, and unique-value count). See the companion script's console log for the extended version including junk-placeholder counts.

| Column                   | Raw dtype | % missing (NaN) | Unique values |
|--------------------------|-----------|-----------------|---------------|
| record_id                | str       | 0.0%            | 48,500        |
| source_system            | str       | 0.0%            | 4             |
| last_updated_ts          | str       | 0.0%            | 43,401        |
| customer_id              | str       | 0.4%            | 28,135        |
| customer_name            | str       | 0.4%            | 3,969         |
| gender                   | str       | 0.3%            | 15            |
| date_of_birth            | str       | 0.4%            | 14,921        |
| age                      | str       | 0.4%            | 68            |
| marital_status           | str       | 0.0%            | 4             |
| residency_status         | str       | 0.4%            | 8             |
| occupation_group         | str       | 0.4%            | 14            |
| annual_income_sgd        | str       | 0.4%            | 38,419        |
| income_band              | str       | 0.4%            | 9             |
| postal_district          | str       | 0.4%            | 32            |
| email                    | str       | 0.4%            | 46,789        |
| mobile_number            | str       | 0.4%            | 47,774        |
| policy_id                | str       | 0.3%            | 47,259        |
| quote_date               | str       | 0.0%            | 1,611         |
| policy_start_date        | str       | 0.0%            | 1,681         |
| product_name             | str       | 0.1%            | 17            |
| product_category         | str       | 0.3%            | 12            |
| distribution_channel     | str       | 0.4%            | 26            |
| payment_frequency        | str       | 0.0%            | 4             |
| annual_premium_sgd       | str       | 0.3%            | 47,814        |
| sum_assured_sgd          | str       | 0.3%            | 3,262         |
| ape_sgd                  | str       | 0.0%            | 47,074        |
| nbp_margin_pct           | str       | 0.3%            | 3,787         |
| policy_term_years        | int64     | 0.0%            | 7             |
| policy_status            | str       | 0.4%            | 9             |
| months_inforce           | str       | 0.4%            | 48            |
| lapse_flag               | str       | 0.0%            | 12            |
| lapse_date               | str       | 90.6%           | 1,009         |
| persistence_13m_flag     | str       | 0.0%            | 12            |
| ip_plan_tier             | str       | 78.5%           | 3             |
| rider_attached_flag      | str       | 0.0%            | 12            |
| rider_premium_before_sgd | float64   | 83.8%           | 7,561         |
| rider_premium_after_sgd  | float64   | 83.8%           | 7,556         |
| copay_pct                | float64   | 83.2%           | 7             |
| deductible_sgd           | float64   | 83.8%           | 5             |
| care_delay_days          | int64     | 0.0%            | 619           |

| Column                      | Raw dtype | % missing (NaN) | Unique values |
|-----------------------------|-----------|-----------------|---------------|
| chronic_condition_flag      | str       | 0.0%            | 12            |
| claim_id                    | str       | 40.2%           | 28,366        |
| claim_date                  | str       | 40.2%           | 1,446         |
| claim_type                  | str       | 40.2%           | 8             |
| claim_amount_sgd            | str       | 40.2%           | 27,866        |
| claim_status                | str       | 40.2%           | 5             |
| claim_decline_reason        | str       | 93.0%           | 8             |
| claim_tat_days              | float64   | 40.2%           | 102           |
| disputed_flag               | str       | 0.0%            | 9             |
| complaint_flag              | str       | 0.0%            | 12            |
| customer_sentiment_score    | float64   | 0.0%            | 197           |
| nps_score                   | int64     | 0.0%            | 11            |
| assessor_id                 | str       | 40.1%           | 207           |
| assessor_approval_limit_sgd | float64   | 40.4%           | 5             |
| approval_above_limit_flag   | str       | 40.2%           | 2             |
| sod_breach_flag             | str       | 40.2%           | 11            |
| fraud_risk_score            | float64   | 40.2%           | 854           |
| agent_id                    | str       | 47.5%           | 1,372         |
| agent_name                  | str       | 47.7%           | 895           |
| agent_tier                  | str       | 47.6%           | 4             |
| agent_tenure_months         | float64   | 47.6%           | 193           |
| agency_unit                 | str       | 47.5%           | 54            |
| agent_status                | str       | 47.6%           | 3             |
| agent_productivity_score    | float64   | 47.6%           | 620           |
| agent_attrition_risk        | float64   | 47.6%           | 510           |

## Appendix B — Companion Artefacts

- **prudential\_eda.py** — the full, reproducible EDA pipeline (structural profiling, quality checks, cleaning, statistics, and all figures in this report).
- **eda\_outputs/cleaned\_dataset.csv** — the standardised, de-duplicated 47,500-row working dataset produced by the script.
- **eda\_outputs/reports/** — descriptive statistics, correlation matrix, categorical-vs-target breakdowns, and the full console log, all in CSV/text form for further analysis.
- **eda\_outputs/figures/** — the complete figure library (univariate, bivariate, multivariate, missing-data), including several views not reproduced in this report for brevity.